



ANALYTICAL REASONING

- You are given two (unmarked) containers of capacity 9 and 4 L and a huge tank of water. Need is to get a measure of exactly 6 L of water. A move is either filling a container completely or emptying a container (either fully or partially). The smallest number of moves needed to do this take is
(a)8 (b)10 (c)12 (d)None
- What is the next letter in the series OTTFFSSSSN?
(a)T (b)O (c)E (d)N
- What is the diameter of the largest circle that can be drawn on a chessboard so that its entire circumference gets covered by the black squares and no part of the circumference on any white space, given that the chessboard has black and white squares of size one inch.
(a)1 inch (b) $\sqrt{2}$ inch
(c) $\sqrt{10}$ inch (d) $2\sqrt{3}$ inch
- A car is filled with $4\frac{1}{2}$ L of fuel for a around trip. If amount of fuel taken while goings is $\frac{1}{4}$ th more than the amount taken for coming, what is the amount of fuel consumed when coming back?
(a)1.5 (b)2 (c)1.75 (d)None
- Which of the following are greater than x when $x = \frac{9}{11}$?
I. $\frac{1}{x}$ II. $\frac{x+1}{x}$ III. $\frac{x+1}{x-1}$
(a)I only (b)I and II only
(c)I and III only (d)II and III only
- Four friends-Arjan, Bhuvan, Guran and Lakha comparing the number of sheep that they owned. It was found that Guran had ten more sheep than Lakha. If Arjan gave one-third to Bhuvan and Bhuvan gave a quarter of what he then held to Guran, who then passed on a fifth of his holding to Lakha, they would all have an equal number of sheep. How many sheep did each of them possess? Give the minimal possible answer.
(a)200, 105, 110, 100 (b)90, 55, 55, 45
(c)180, 110, 110, 100 (d)None of these
- In a class, six students P, Q, R, S, T and U are the top six rank holders, not necessarily in the same order. R did not get the 4th rank. P's rank is higher than U's and R's but lower than Q's. Among these six rankers, there are four students whose ranks are lower than S's rank and five students whose ranks are above that of T. who is ranked 5th in the class?
(a)U (b)T (c)R (d)None
- Three players-Aalu, Kachaalu and Bhalu were playing poocker and suddenly started to quarrel among themselves blaming each other for cheating. It was found out that atleast one person among the three cheated. When they were asked who cheated, their replies were as follows.
Aalu I did not cheat, Kachaalu cheated
Kachaalu I did not cheat, Both Aalu and Bhalu cheated.
Bhalu I did not cheat, only Kachaalu did not cheat.
If exactly one person among them always spoke truth, another always lied and the third alternated between the truth and lie, then which of the following statements can never be true is any case?
(a) Only Aalu and Bhalu cheated
(b) Only Aalu and Bhalu did not cheat
(c) Bhalu always spoke the truth
(d) Bhalu alternated between truth and lie.
- If x and y are the two digits of the number 565xy such that this number is divisible by 80, then $x + y$ is equal to
(a)2 (b)3 (c)8 (d)6
- If both 7^2 and 3^3 are factors of the number $(a \cdot 11^3 \cdot 6^2 \cdot 3^{11})$, then what is the smallest possible value of a ?

- (a)1323 (b)147 (c)21 (d)3087
- Let x , y and z be distinct integers, x and y are odd and positive, and z is even and positive. Which one of the following statements cannot true?
(a) $(x - z)^2 y$ is even (b) $(x - z) y^2$ is odd
(c) $(x - z) y$ is odd (d) $(x - y)^2 z$ is even
- Form a height of 16 m, a ball fell down and each time it bounces half the distance back. What is the distance travelled?
(a)45 m (b) ∞ (c)48 m (d)24 m
- If a man walks at the rate of 4 km/h, he misses a train by only 6 min. However, if he walks at the rate of 5 km/h he reaches the station 6 min before the arrival of the train. Find the distance covered by him to reach the station.
(a)4 km (b)7 km (c)9 km (d)5 km

Direction (Q. Nos. 14-17) Read the following statements and answer questions.

The office staff of XYZ corporation presently consists of three book keepers, P, Q, R and 5 secretaries S, T, U, V, W. the management is planning to open a new office in another city using 2 book keepers and 3 secretaries of the present staff. To do so they plan to separate certain individuals who don't function well together. The following guidelines were established to set up the new office.

- Book keepers P and R are constantly finding fault with one another and should not be sent together to the new office as a team.
 - R and T function well alone but as a team, they should be separated.
 - S and V have not been on speaking terms and shouldn't go together.
 - Since, S and U have been competing for promotion they shouldn't be a team
- If P is to be moved as one of the bookkeepers, which of the the following cannot be a possible working unit?
(a)PQSTW (b)PQSVW
(c)PQTUW (d)PQTVW
 - If R and U are moved to the new office, how many combinations are possible?
(a)1 (b)2 (c)3 (d)4
 - If R is sent to the new office, which member of the staff cannot go with R?
(a)Q (b)S (c)W (d)V
 - If S goes to the new office, which of the following is true?
(a)Only R cannot go (b)Only P cannot go
(c)Only P and R cannot go (d)R cannot go and W must go
 - Substitute digits for the letters to make the following relation true.
$$\begin{array}{r} S T I L L \\ + W I T H I N \\ \hline L I M I T S \end{array}$$

Note: The leftmost letter can't be zero in any word. Also, there must be a one-to-one mapping between digits and letters. e.g., if you substitute 3 for the letters S, no other letter can be 3 and all other S in the puzzle must be 3.
(a)98533 + 258056 = 356589
(b)41211 + 527013 = 938224
(c)98533 + 158056 = 256589
(d)47166 + 517013 = 614179
 - 12 members were present at a board meeting. Each member shook hands will all of the other memebbers before and after the meeting. How many hand shakes were there?
(a)118 (b)127 (c)132 (d)264
 - The letters P, Q, R, S, T, U and V not necessarily in that order represents seven consecutive integers from 22 to 33.





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U is as much less than Q as R is greater than S.

V is greater than U.

Q is the middle term.

P is 3 greater than S.

Can you find the sequence of letters from the lowest. Value to the highest value?

(a)PVSQRTU (b)SUTQPRV

(c)USVQPRT (d)TUSQPRV

21. There were a total of 10 bicycles and tricycles. If the total number of wheels was 24, how many tricycles were there?

(a)4 (b)6 (c)8 (d)2

22. A person travels on a cycle from home to church on a straight road with wind against him. He took 4 h to reach there. On the way back to the home, he took 3 h to reach as wind was in the same direction. If there is no wind, how much times does he take to travel from home to church?

(a)3 h 35 min 12 s (b)3 h 32 min 32 s

(c)3 h 30 min 00 s (d)3 h 25 min 42 s

23. What are the next three numbers in the given series

1 1 2 1 2 2 3 1 2 2 3 2 3 3 4 1 2 2 3 2 3 3 4 2 3 3?

(a)2, 3, 4 (b)2, 3, 2 (c)1, 2, 3 (d)4, 3, 4

24. In the middle of the confounded desert, there is the lost city of "Ash". To reach it, I will have to travel overland by foot from the coast. On a trek like this, each person can only carry enough rations for five days and the farthest we can travel in one day is 30 miles. Also, the city is 120 miles from the starting point. What I am trying to figure out is the fewest number of persons, including myself, that I will need in our group so that I can reach the city, stay overnight, and then return to the coast, without running out of supplies. How many persons (including myself) will I need to accomplish this mission?

(a)5 (b)6 (c)4 (d)3

25. A woman took a certain number of eggs to the market and sold some of them. The next day through her poultry industry, the number left over had been doubled, and she sold the same number as the previous day. On the third day, the new remainder was tripled, and she sold the same number as before. On the fourth day, the remainder was quadrupled, and her sale were the same as before. On the fifth day, what had been left over were quintupled, yet she sold exactly the same as on all the previous occasions and so disposed of her entire stock. What is the smallest number of eggs she could have taken to the market the first day, and how many did she sell daily?

(a)110.50 (b)127.65

(c)100.60 (d)103.60

26. The Bulls, Pacers, Lakers and Jazz ran for a contest. Anup, Sujit, John made the following statements regarding results.

- Anup said either Bulls or Jazz will definitely win.
- Sujit said he is confident that Bulls will not win.
- John said he is confident that neither Jazz nor Lakers will win.

When the results came, it was found that only one of the above three had made a correct statement. Who has made the correct statement and who has won the contest?

(a) Anup, Bulls (b) John, Pacers

(c) Sujit, Lakers (d) Sujit, Jazz

27. A certain street has 1000 buildings. A sign-maker is contracted to number the houses from 1 to 1000. How many zeroes will be needed?

(a)128 (b)190 (c)181 (d)192

28. Examine the following sequence of numbers

1

1 1

2 1

1 2 1 1

1 1 1 2 2 1

3 1 2 2 1 1

1 3 1 1 2 2 2 1

1 1 1 3 2 1 3 2 1 1

3 1 1 3 1 2 1 1 1 3 1 2 2 1

What are the next two numbers in the given series?

(a)13211311122111231131 and 11213211321222111131221133

(b)23113112211132113111 and 11121321132122211131221133

(c)11231131122111321131 and 11131221212221133112132113

(d)13211311123113112211 and 1113122113311213211321221

29. There were two men standing on a street. The one says to the other. "I have 3 daughters, the product of their ages is 36. What is the age of the OLDEST daughter?"

The second guy says, "I need more information. "So, the first guy says, "The sum of their ages is equal to the address of the house across the street."

The second guy looks at the address and says, "I still need more information". So, the first guy says. "My oldest daughter wears a red dress."

(a)9 (b)6 (c)12 (d)4

30. Three Gold (G) coins, three Silver (S) coins and three Copper (C) coins are arranged in a single row as follows

- Only 2 adjacent unlike coins can be moved at any one time.
- The moved coins must be in contact with atleast one other coin in line i.e., no pair of coins is to be moved and placed away from the remaining ones.
- No coin pairs can be reversed i.e., S-C combination must remain in that order in its new positions when it is moved.

What is the minimum number of moves required to get all the coins in following order?

C C C S S S G G G

(a)6 (b)9 (c)8 (d)12

31. Mr. and Mrs. Birla and Mr. and Mrs. Tata competed in a chess tournament. Of the three games played

(I) In only the first game were the two players married to each other.

(II) The men won two games and the women won one game.

(III) The Birlas won more games than the Tatas.

(IV) Anyone who lost a game did not play a subsequent game.

Who did not lose a game?

(a)Mr Birla (b)Mrs Birla

(c)Mr Tata (d)Mrs Tata

32. Of the three numbers, second is twice the first and is also thrice the third. If the average of three numbers is 44, the largest number is

(a)24 (b)36 (c)72 (d)108

33. Larger, medium and small ships are used to bring water. 4 large ships carry as much water as 7 small ships. 3 medium ships carry the same amount of water as 2 larger ships and 1 small ship. 15 large, 7 medium and 14 small ships, each made 36 journeys and brought a certain quantity of water. In how many journeys would 12 large, 14 medium and 21 small ships bring the same quantity of water?

(a)32 (b)25 (c)29 (d)49

34. Five men P, Q, R, S and T read newspaper. The one who reads first gives it to R. the one who reads last had taken it from P. T was not the first or the last to read. There were two readers between Q and P. To whom did Q pass the newspaper?

(a)R (b)P (c)S (d)T





35. An airline has a certain free luggage allowance and charges for excess luggage at a fixed rate per kg. Two passengers Raja and Rahim have 60 kg luggage between them, and are charged ₹ 1200 and ₹ 2400, respectively for excess luggage. Had the entire luggage belonged to one of them, the excess luggage charge would have been ₹ 5400. What is the weight of Rahim's luggage?

(a) 20 kg (b) 25 kg (c) 30 kg (d) 35 kg

36. A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. What numbers of rows is not possible?

(a) 3 (b) 4 (c) 5 (d) 6

Directions (Q. Nos. 37-40) Read the following passage and answer the questions.

Sports (and game) persons P, Q, R, S, T, U and V of a university are at the Bengaluru Airport. Five of them are selected players and leaving to participate in the grand sports Event in five different events cricket, carrom, badminton and table tennis being held at 5 different cities Mumbai, Chennai, Kolkata, Delhi and Hyderabad.

- (i) P is going to Delhi, but he does not play either cricket or carrom
(ii) Q has come to give send off to R, who is a chess player and is not leaving to either Mumbai or Hyderabad.
(iii) S is leaving to Kolkata to play table tennis.
(iv) T is not a selected player.

37. Who plays badminton?

(a) P (b) Q (c) R (d) S

38. Cricketer goes to

(a) Mumbai (b) Hyderabad
(c) Chennai (d) Delhi

39. Player of which game goes to Delhi?

(a) Badminton (b) Chess
(c) Cricket (d) Table tennis

40. Who plays chess and where is he going?

(a) R and Chennai (b) S and Mumbai
(c) U and Delhi (d) None of these

Directions (Q. Nos. 41-45) Read the following information carefully and answer the questions.

- (i) $P \psi Q$ means P is mother of Q
(ii) $P \in Q$ means P is sister of Q
(iii) $P \$ Q$ means P is father of Q
(iv) $P \# Q$ means P is brother of Q

41. Which of the following means N is definitely daughter of K?

(a) $K \$ L \# M \# N$ (b) $M \psi K \$ N \in L$
(c) $K \psi M \# L \in N$ (d) $L \psi K \$ N \# M$

42. Which of the following means R is brother of T?

(a) $R \psi S \# U \$ T$ (b) $U \psi R \# S \# T$
(c) $U \psi R \in S \psi T$ (d) $T \# \$ \$ Q \in R$

43. Which of the following means X is real grandmother of Y?

(a) $X \in Z \psi K \$ L \# Y$ (b) $Y \psi K \$ X \# L$
(c) $Y \# L \$ K \psi X \in Z$ (d) $K \# X \psi Z \# L \$ Y$

44. If $K \psi L \in M \# N$, then how K is related with N?

(a) Mother (b) Aunt
(c) Great Aunt (d) Grandmother

45. Which of the following means K is related K is nephew of M?

(a) $N \# M \$ L \# K \in O$ (b) $K \# L \$ N \in O \$ M$
(c) $L \psi O \# M \$ O \in K$ (d) $M \# N \$ L \# K \$ O$

46. There are six houses in a row. Mr. Lal has Mr. Babu and Mr. Anil as neighbours. Mr. Bhatia has Mr. Gupta and Mr. Sharma as neighbours. Mr. Gupta's houses is no next to Mr. Babu or Mr. Anil and Mr. Sharma does not live next to Mr. Anil. Who are Mr. Babu's next-door neighbours?

- (a) Mr. Lal and Mr. Bhatia (b) Mr. Lal and Mr. Anil
(c) Mr. Sharma and Mr. Lal (d) Only Mr. Lal

47. A watch which gains 10 s in 5 min was set correct at 9 am when the watch indicated 20 min past 7 O' clock, the same evening, the true time is

(a) 7 pm (b) 7 : 40 pm (c) 7 : 10 pm (d) 8 pm

48. A boy observes the reflection of a clock in a mirror. The time observed by the boy in the mirror is 3 h 45 min. what is the actual time shown in the clock?

(a) 8 h 45 min (b) 9 h 45 min
(c) 8 h 15 min (d) 9 h 15 min

49. Gold is 19 time as heavy as water and copper is 9 times as heavy as water. In what ratio should these be mixed to get an alloy 15 times as heavy as water?

(a) 1 : 2 (b) 2 : 3 (c) 3 : 2 (d) 1 : 1

50. In an objective type examination, 120 objective type questions are there : each with 4 options P, Q, R and S. A candidate can choose either one of these options or can leave the question unanswered.

How many different ways exist for answering this question paper?

(a) 5^{120} (b) 4^{120} (c) 120^5 (d) 120^4

ANSWER

1.	A	2.	D	3.	B	4.	B	5.	B
6.	D	7.	C	8.	D	9.	D	10.	B
11.	A	12.	C	13.	A	14.	B	15.	A
16.	B	17.	D	18.	A	19.	C	20.	D
21.	A	22.	D	23.	D	24.	C	25.	D
26.	C	27.	D	28.	A	29.	A	30.	A
31.	A	32.	C	33.	C	34.	A	35.	D
36.	D	37.	A	38.	B	39.	A	40.	A
41.	B	42.	B	43.	D	44.	A	45.	D
46.	C	47.	A	48.	C	49.	C	50.	A

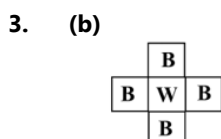




SOLUTION

1. (a)
Step 1 Fill; 9 L container.
Step 2 Pour 4 L water from 9 L container to 4 L container.
Step 3 Empty 4 L container.
Step 4 Again pour 4 L water from 9 L container to 4 L container.
Step 5 Again empty 4 L container.
Step 6 Now pour remaining 1 L water from 9 L container to 4 L container.
Step 7 Fill 9 L container
Step 8 Pour water from 9 L container to 4 L container, so that it is completely full.
Now it is completely full.
Now, 9 L container contains 6 L water.

2. (d)
The letters are repeat ONE, TWO, THREE, FOUR and FIVE times respectively.
So, the required letter is N.



The required diameter will be diagonal of white square which will give it's middle point as the centre.
So, diameter = $\sqrt{2}$ inch.

4. (b) If x is the amount of fuel consumed while coming back.
Then $x + x + \frac{x}{4} = 4\frac{1}{2}$
 $\Rightarrow \frac{9x}{4} = \frac{9}{2}$
 $x = 2$

5. (b)
At $x = \frac{9}{11}$; $\frac{1}{x} = \frac{11}{9} > x$:
At $x = \frac{9}{11}$; $\frac{x+1}{x} = \frac{20}{9} > x$
At $x = \frac{9}{11}$; $\frac{x+1}{x-1} = 10 < x$
Hence, I and II are correct.

6. (d) None of the given answer is correct from the available data set.
 $G = L + 10$
 $\frac{2A}{3} = \frac{3}{4} \left(B + \frac{A}{B} \right) = \frac{4}{5} \left(G + \frac{1}{4} \left(B + \frac{A}{3} \right) \right)$
 $= L + \frac{1}{5} \left(G + \frac{1}{4} \left(B + \frac{A}{3} \right) \right)$
 $\Rightarrow \frac{2A}{3} = \frac{3B}{4} + \frac{A}{4} = \frac{4G}{5} + \frac{B}{5} + \frac{A}{15}$
 $= L + \frac{G}{5} + \frac{B}{20} + \frac{A}{60}$
 $\Rightarrow A = 180; B = 100; G = 110; L = 80$
Which become inconsistent.

7. (c)
- 
- Rank from data, top is bottom
Is given.
Hence, R's rank is 5th in the class.

8. (d)
If Bhalu alternated between truth and lie, so will Kachaalu, which will be a contradiction, as only one person alternated between truth and lie.

9. (d)
 $565xy$ will be divisible by 80, if $x = 6$ and $y = 0$
 $\Rightarrow x + y = 6$

10. (b)
 7^2 and 3^3 will be factor of
(a. $11^3 \cdot 6^3 \cdot 13^{11}$) If $a = 7^2 \cdot 3 = 147$.

11. (a)
Odd-even = Odd, so $(x - z)^2 y$ will be odd. Hence, (a) cannot be true.

12. (c)
Required distance
 $= 16 + 2 \left[16 \times \frac{1}{2} + 16 \times \frac{1}{2^2} + 16 \times \frac{1}{2^3} + 16 \times \frac{1}{2^4} + \dots \right]$
 $= 16 + 2 \frac{16 \times \frac{1}{2}}{1 - \frac{1}{2}} = 48 \text{ m}$

13. (a)
If d is the distance covered, then if 't' hour is the required time to catch the train in time, then

$$\frac{d}{4} = t + \frac{1}{10}$$

$$\text{and } \frac{d}{5} = t = \frac{1}{10}$$

$$\Rightarrow \frac{d}{20} = \frac{1}{5}$$

$$\Rightarrow d = \frac{20}{5} = 4 \text{ km}$$

14. (b) PQSVW is not possible, as S and V cannot be together.

15. (a)
If R is in new office, then as book keeper there is only one option Q. Due to R and U in new office T and S cannot be considered. So, V and W will be sent as new secretaries. So, there is only one option QRUVW.

16. (b)
If R goes, T won't go and if S went further U and V won't go, so we will not get three secretaries.

17. (d)
If S goes to new office, then team will be S, T, W, P, Q only. So, R cannot go and W must go.

18. (a)
- | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---------------|---|---|---|---|---|---|
| | S | T | I | L | L | | 9 | 8 | 5 | 3 | 3 | | |
| + | W | I | T | H | I | N | $\Rightarrow$ | 2 | 5 | 8 | 0 | 5 | 6 |
| | L | I | M | I | T | S | | 3 | 5 | 6 | 5 | 8 | 9 |

19. (c)
Total number of hand shakes will be $2^{(12C_2)} = 132$

20. (d)
From given information's, we get
 $Q - U - R - S$;
 $V > U$; $P = S + 3$



⇒ Sequence from the lowest to the highest

I U S Q R P V

21. (a)

If x tricycles are there, then there will be $(10 - x)$ bicycles.

$$3x + 2(10 - x) = 24$$

$$x + 20 = 24$$

$$\Rightarrow x = 4$$

22. (d)

Let the speed of person and that of wind be u and v km/h respectively. d

$$\text{Then } \frac{d}{u-v} = 4$$

$$\text{and } \frac{d}{u+v} = 4$$

$$\Rightarrow 4(u - v) = 3(u + v)$$

$$\Rightarrow u = 7v$$

$$\Rightarrow \frac{d}{u + \frac{u}{7}} = 3$$

$$\Rightarrow \frac{d}{\left(\frac{8u}{7}\right)} = 3$$

$$\Rightarrow \frac{d}{u} = \frac{24}{7}$$

$$\text{Hence, required time} = \frac{24}{7} \text{ h} = 3 \text{ h } 25 \text{ min } 42 \text{ sec}$$

23. (d)

Each 1 in the given series has the digits on its right exceeding by the digits on the left by

1 1 2; 1 1 2 1 2 2 3;

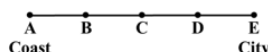
1 1 2 1 2 2 3 1 2 2 3 2 3 3 4

1 1 2 1 2 2 3 1 2 2 3 2 3 3 4 1 2 2 3 2 3 3 4 2

3 3 4 3 4 4 5

So, next three numbers is 4, 3, 4.

24. (c)



The person himself reach the city E from coast A, by travelling through B, C and D each at 30 km each in 4 days.

Then, when returning he will spend his rations at D, where he will be joined by second person with ration of 2 days. At C, third person will be there with ration of 3 persons for one day. Again at B, fourth person will be there, with ration of 4 persons for one day, and hence finally all 4 will reach the coast A.

25. (d)

If y was the total number of eggs and x is the daily sale. Then,

$$\text{Left over egg on day 1} = y - x$$

$$\text{Left over egg on day 2} = 2(y - x) - x = 2y - 3x$$

$$\text{Left over egg on day 3} = 3(2y - 3x) - x$$

$$= 6y - 10x$$

$$\text{Left over egg on day 4} = 4(6y - 10x) - x$$

$$= 24y - 41x$$

$$\text{Left over egg or say total eggs on day 5}$$

$$= 5(24y - 41x) = x$$

$$\Rightarrow 120y - 205 = x$$

$$\Rightarrow 120y = 206x$$

$$\Rightarrow 60y = 103x$$

$$\Rightarrow y = 103 \text{ and } x = 60 \text{ as minimum values.}$$

26. (c) Sujit has told the truth and Lakers will be the winner, so automatically remaining two statements become wrong.

27. (d)

0 9 Case for two digit numbers.

$\bar{1}$

0

$$\bar{1} - \bar{1} \quad 9 \times 10 = 0 \text{ cases}$$

9 For three digit numbers.

0

$$\bar{1}\bar{1} \quad 9 \times 10 = 0 \text{ cases}$$

In 1000 there is 3 O's

$$\text{Total O's } 9 + 90 + 90 + 3 = 192$$

28. (a)

There is continuation of series.

111 211 211 111 221 312 211 131 122 211 113 213
211 311 312 111 312 211 321 131 112 211 123 113
111 213 211 321 222 111 131 221 133

29. (a)

From the first information oldest daughter will be 9 or 12 yr old.

From the next information her age will be 9.

$$(AS - 9 + 4 = 13)$$

30. (a)

Q S C G S C G S C → C G S G S C G S C
→ C G S C G S C S G → C C G G S S C S G
→ C C C S G G S S G → C C C S S G G G S
→ C C C S S G S S G → C C C S S S G G G

Seven moves are shown here out of which one of them is trivial. i.e., six moves are required.

31. (a)

1st Game Mr. Birla defeated Mrs. Birla

2nd Game Mr. Birla defeated Mrs. Tata

3rd Game Mrs. Birla defeated Mr. Tata

32. (c)

Second number is the largest. Let it be x, then first number is $x/2$ and third number is $x/3$.

So, their average is

$$\frac{x + \frac{x}{2} + \frac{x}{3}}{3} = 44$$

$$\Rightarrow \frac{11x}{18} = 44$$

$$\Rightarrow x = \frac{44 \times 18}{11} = 72$$

33. (c) 15 large, 7 medium and 14 small ships

$$= 15 \times \frac{(2 \times \frac{7}{4} + 1)}{3} + 14 \text{ small ships.}$$

$$= \frac{105}{4} + \frac{21}{2} + 14 = \frac{203}{4} \text{ small ships.}$$

Now, 12 large, 14 medium and 21 small ships

$$= 12 \times \frac{7}{4} + \frac{14(2 \times \frac{7}{4} + 1)}{3} + 21 \text{ small ships}$$

$$= 21 + 21 + 21 = 63 \text{ small ships}$$

$$\text{Number of trips by } \frac{203}{4} \text{ ships} = 36$$

$$\text{So, number of trips by 63 ships} = \frac{203}{4} \times \frac{36}{4} = 29 \text{ trips.}$$

34. (a) Q R I P S

According to the given information the order of reading newspaper, so Q passed newspaper to R.

35. (d)

If free luggage is y kg, then Raja has x kg, and Rahim has 2x kg extra luggage.

$$\text{Then, } 2y + 3x = 60$$





If $3x$ kg extra luggage cost 3600, then 5400 will be cost of

$$\frac{5400}{3600} \times 3x = \frac{9}{2}x$$

$$\Rightarrow y + 3x = \frac{9}{2}x$$

$$\Rightarrow y = \frac{3}{2}x$$

$$\Rightarrow x = 10 \text{ kg}$$

and $y = 15 \text{ kg}$

So, Rahim's luggage is $y + 2x = 35 \text{ kg}$

36. (d)

If x is the number of rows, then if number of children in last row is y , then

$$y + (y + 3) + (y + 6) + \dots + \{y + 3(x - 1)\} = 630$$

$$\Rightarrow xy + \frac{3(x-1)}{2} \cdot 630$$

$$x = 3 \Rightarrow 3y + 9 = 630 \text{ (possible, } y = 207)$$

$$x = 4 \Rightarrow 4y + 18 = 630 \text{ (possible, } y = 153)$$

$$x = 5 \Rightarrow 5y + 30 = 630 \text{ (possible, } y = 120)$$

$$x = 6 \Rightarrow 6y + 45 = 630 \text{ (impossible)}$$

Solutions (Q. Nos. 37-40)

City / Player	M	C	K	D	H
P	x	x	x	✓	x
R	x	✓	x	x	x
S	x	x	✓	x	x
U	✓	x	x	x	x
V	x	x	x	x	✓

Game / Player	Cr.	Ch.	Ca.	Ba.	T.T.
P	x	x	x	✓	x
R	x	✓	x	x	x
S	x	x	x	x	✓
U	x	x	✓	x	x
V	✓	x	x	x	✓

37. (a) P

38. (b) Hyderabad

39. (a) Badminton

40. (a) R and Chennai

41. (b) $M \Psi K \$ N \in L$

$\Rightarrow M$ is mother of K , who is father of N , who is sister of L .

Hence, N is daughter of K .

42. (b)

$U \Psi R \# S \# T$

$\Rightarrow R$ is brother of S who is brother of T .

Hence, R is brother of T .

43. (d) $K \# X \Psi Z \# L \$ Y$

$\Rightarrow X$ is mother of Z , who is brother of father of Y .

Hence, X is real grandmother of Y .

44. (a) $K \Psi L \in M \# N$

$\Rightarrow K$ is mother of sister of brother of N .

Hence, K is mother of N .

45. (d)

$M \# N \$ L \# K \$ O$

$\Rightarrow M$ is brother of father of brother of K who is father of O .

Hence, K is nephew of M .

46. (c)

From the given data following is the possibility.

Anil Lal Babu Sharma Bhatia Gupta

$\Rightarrow$ Babu has Lal and Sharma as the neighbours.

47. (a)

As the watch gains 10 s in 5 min so it gains 2 min in 1 h. 20 min past 7 O'clock implies 10 h 20 min after 9 am, so actual time is 10 h ahead and 20 min is the gain. Hence, actual time is 7 pm.

48. (c)

As by reflexion left hand becomes right hand and vice-versa, so in original clock hour hand is between 8 and

49. (c)

Let the ratio be $\lambda : 1$

then, $19\lambda + 9 = (1 + \lambda)$

$$\Rightarrow 4\lambda = 6$$

$$\Rightarrow \lambda = \frac{6}{4}$$

$$\Rightarrow \lambda : 1 :: 3 : 2$$

50. (a)

Each question has 5 options, as it can be answered as A B, C, D or left also.

Now, all 120 questions will have 5 independent options.

Hence, total number of required ways

$$= 5 \times 5 \times 5 \dots \times 5 \text{ (120 times)} = 5^{120}.$$

